

AnyCam 3.8.0-rc1.0 Audit Report

Release audit - 10 October 2026 - B54 - tag 3.8.0-rc1.0

Scope confirmed by CrystalHeeler on 2026-10-10: B54 only, in a new minor version. The design was agreed point by point on 2026-10-09 and drawn in docs/B54_Flow_Comparison.html (right side). Four research notes support it: docs/B54-Port-Probe-Research.md and docs/B54-Followup-Research.md. Not field-tested.

What B54 changes

Area	Before	Now
Scan, ONVIF	ONVIF stream paths read only at password entry; the ONVIF pass fetched a web page, an RTSP OPTIONS and a no-login walk on 554	A device that answered WS-Discovery gets one GetProfiles with no login. A login prompt gives a "needs password" card and no further probing; profiles give a ready card when a stream plays without a login.
Scan, RTSP walk	Brand paths, then the universal list, in one pass; the walk stopped only after 5 same-realm 401s (which the Microseven never sent)	Brand paths first; the first 401 stops the walk (a password is needed); the full list only for a camera with no ONVIF and no brand match.
Scan, other ports	Each extra web port drew 17 MJPEG, 11 HLS, 9 WebRTC and 7 WebSocket requests	Once ONVIF or a brand path has identified the camera, its other ports are not probed.
Scan, Layer 2	A multi-socket walk ran when Layer 1 heard RTSP but found nothing	No Layer 2 at scan time (no documented camera needs it); Layer 1 reopens one connection on "Connection: close", reads replies strictly, and sends a User-Agent.
Password entry	ONVIF, then always a brand-path pass after the walk	The 0/1/2 rule: 2 streams from ONVIF stop; 1 checks the brand paths; 0 runs the full walk. Each walk stops at 2 streams.
Login attempts	Every 401 path got a login attempt (a typo could lock a camera out)	The walk stops at the first rejected login; after one success a later 401 skips that path, at most 2. The card says "Password rejected or camera locked".
DVR channels	Each channel's main stream only	Each channel, and the parent, also gets its sub-stream checked.
Deep Re-Probe	Resumed an early-stopped walk, sometimes Layer 2	One full walk of every path, Layers 1 and 2 (also on skip_layer2 brands).
Brand table	Hikvision NVR lacked skip_layer2 its comment claimed; the Axis retry hung on an unreachable throttle_type	Hikvision NVR gets skip_layer2; the Axis retry is its own field (rtsp_query_retry) and runs.

Tests run

Suite	Result	Scope
Release gate	PASS	Eight gates.

Suite	Result	Scope
Python integration	PASS	673 of 673; new section AP (7 checks) drives the walker against a fake RTSP server: scan 401-stop, the 0/1/2 rule, the login rule, and the Connection: close reopen. Sections U and Z updated.
JavaScript behaviour	PASS	193 of 193.
Not run	NOTED	No field test. No camera exercised the new scan on real hardware; the Microseven is still stuck pending a power cycle.

Best-practices compliance

Measured against *docs/AnyCam_Coding_Best_Practices.md*. Rows this change touches.

Standard	Check applied	Result	Note
1.12 Logging	Truth	PASS	Each stop logs its reason: "password needed", "identified on port N", "login rejected".
6.1 Security	No secrets at rest	PASS	No password in a logged URL; the lock hint request sends no login.
1.18 Testing	New tests first	PASS	Section AP added with the walker rewrite; U and Z updated in the same change.
1.2 Runtime errors	Socket handling	PASS	The reopen caps at 3; the reply reader keeps leftover bytes apart.
Rule 9 (privacy)	New text	PASS	Made-up addresses only.

Known issues

- **Microseven:** stuck until power-cycled; then the new scan and the ffmpeg copy get their test.
- **B48:** go2rtc's local RTSP server does not ask local programs for a password.
- **B33:** one Amcrest camera can get two cards.
- **Later (B55–B59):** the MAC-first check and printer rule, Reolink FLV / Synology / rpos, the ONVIF NVR split, the brand-table third stream, and the Reolink hub timeout.

Files changed

File	Change
anycam_probe.py	The walker: scan 401-stop, the login rule, want_streams, the Connection: close reopen, a strict reply reader, a User-Agent; find_rtsp_path: scan/deep/want_streams/path_set, no Layer 2 at scan time, rtsp_query_retry; onvif_probe_no_login.
anycam_scan.py	ONVIF first with no login; stop probing a host's other ports once identified; the ONVIF pass uses the no-login GetProfiles.

File	Change
anycam_credentials.py	The 0/1/2 rule; the sub-stream from the walk; the login-rejected message and the lock hint; DVR sub-streams per channel; Deep Re-Probe is one full walk.
camera_db.py	Hikvision NVR skip_layer2; Axis rtsp_query_retry.
verify_release.py	find_rtsp_path contract updated for B54.
tests/	Section AP; sections U and Z updated.
CHANGELOG.md, config.yaml, docs	3.8.0-rc1.0; build plan.